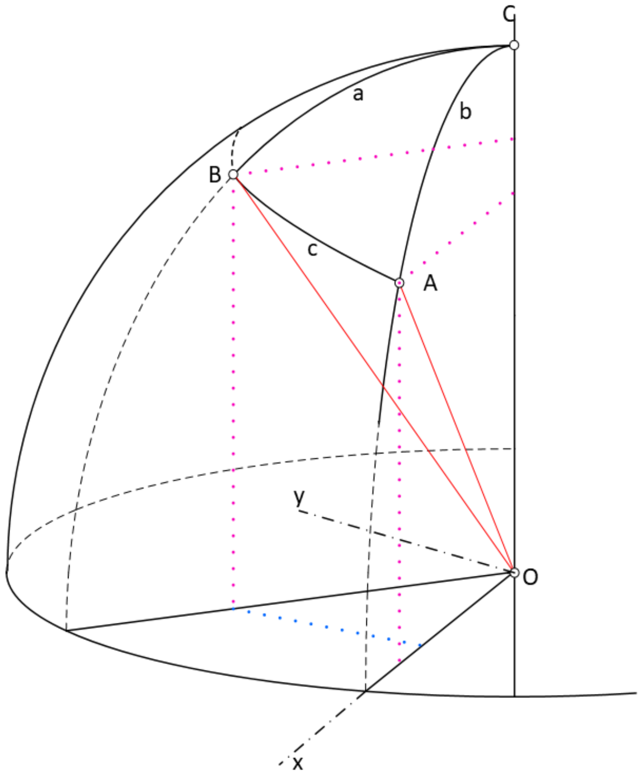
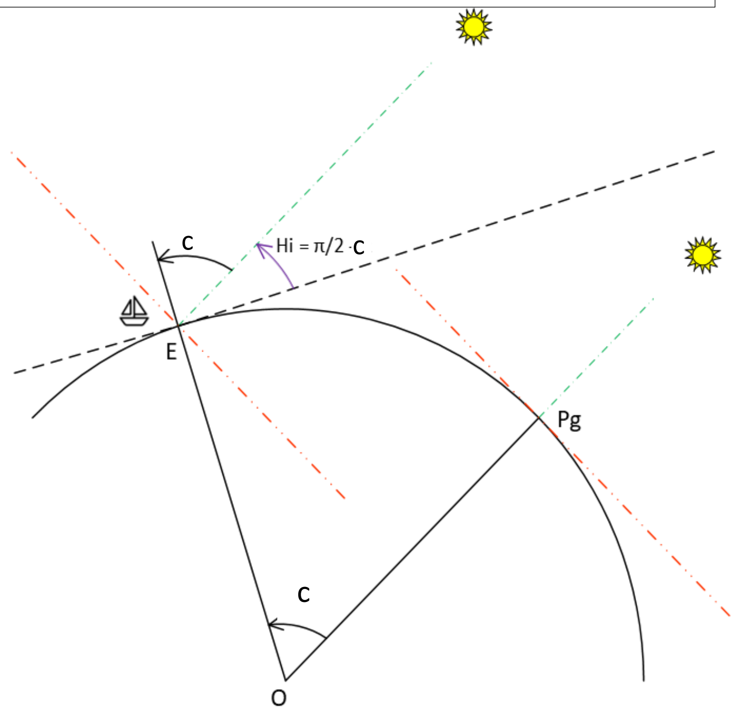
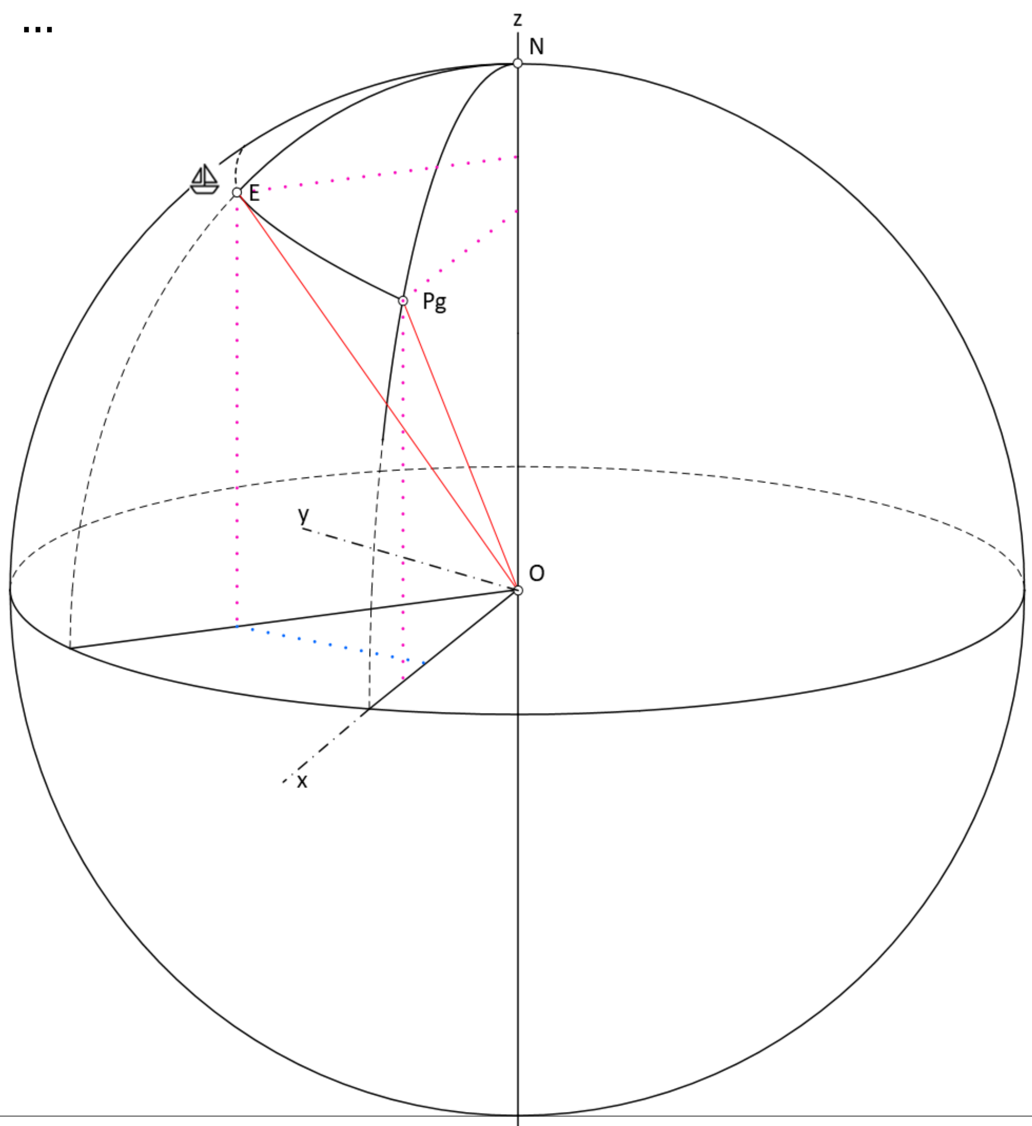
$$Hc = \sin^{-1}(\sin(Latitude_E) \times \sin(Declination) + \cos(Latitude_E) \times \cos(Declination) \times \cos(LHA))$$





$$\overrightarrow{OA} * \overrightarrow{OB} = \ldots$$



$$Z = \cos^{-1}((\sin(Declination) - \sin(Latitude_E) \times \sin(H_c)) / (\cos(Latitude_E) \times \cos(H_c)))$$